Apex Systems

Response to RFP NW-2025-014: Vendor Onboarding Platform for Northwind Logistics

Prepared for Northwind Logistics - submitted 3 March 2025 - valid for 90 days

1. Executive Summary

Northwind Logistics needs one platform to onboard, verify and continuously monitor roughly 3,000 vendors across freight, warehousing and last-mile delivery. Apex Systems proposes a multi-tenant, API-first onboarding platform on a microservices architecture, deployed across two cloud regions with automatic failover.

Our design puts security, auditability and integration depth first. We will deliver a production-ready platform in 28 weeks for a fixed fee. The price reflects a premium engineering and security investment, and we explain below what it buys.

2. Our Understanding of Northwind's Requirements

From the RFP and the clarification session we understand that Northwind must:

- cut vendor onboarding from the current 12 working days to under 5;
- automate sanctions, tax-ID and bank-detail verification with a full audit trail;
- synchronise approved vendors with SAP S/4HANA and Oracle Transportation Management;
- meet internal audit expectations for access control and data retention.

3. Solution Architecture

The platform is event-driven. Stateless services scale horizontally behind an API gateway, a managed message broker decouples workflow steps, and a PostgreSQL cluster with read replicas stores vendor records. Load tests on a comparable deployment sustained 1,200 requests per second with a 95th-percentile latency under 180 ms.

Table 1 - Core components

Component	Technology	Purpose
API gateway	Managed gateway, OAuth 2.0	Authentication, rate limiting
Workflow engine	Stateless services, message broker	Onboarding steps and approvals
Data store	PostgreSQL cluster, read replicas	Vendor records, audit trail
Document service	Object storage, content hashing	Contracts and certificates
Observability	Central logging and tracing	Monitoring and alerting

4. Integration Approach

Integrations use a documented REST and webhook API plus certified connectors for SAP S/4HANA, Oracle Transportation Management and Microsoft Entra ID. Each connector ships with contract tests and a sandbox so Northwind can verify behaviour before go-live.

- Inbound: vendor master data and payment terms from SAP;

- Outbound: approved-vendor events to Oracle TM within 60 seconds;
- Identity: SAML 2.0 and OIDC single sign-on with enforced MFA.

5. Security and Compliance Framework

Apex Systems holds SOC 2 Type II and ISO 27001 certifications, both audited in the last 12 months. Data is encrypted in transit with TLS 1.3 and at rest with AES-256 using customer-managed keys.

Table 2 - Assurance status

Standard	Status	Last audit
SOC 2 Type II	Certified	November 2024
ISO 27001	Certified	January 2025
GDPR data processing agreement	Available	Standard terms

All administrative and data-access events are written to an immutable audit log retained for seven years. An independent penetration test is performed before go-live and annually. Principal risks are legacy ERP complexity, mitigated by early connector spikes, and data-migration quality, mitigated by two dry-run migrations.

6. Hosting, Resilience and Operations

The platform is hosted in two cloud regions in an active-passive arrangement. Databases are backed up daily with 35-day retention, and backups are restored into an isolated environment every quarter to prove they work. Releases follow a blue-green pattern so that deployments do not interrupt vendors or buyers.

Table 5 - Service objectives

Objective	Target	Measured by
Availability	99.95% monthly	Synthetic transaction checks
API latency (p95)	Under 250 ms	Gateway metrics
Planned maintenance	Outside 06:00-20:00 UTC	Change calendar

7. Data Migration and Testing Strategy

Vendor data will be migrated from three sources: the current ERP vendor master, procurement spreadsheets and the document share. We profile each source in discovery, agree matching rules with Northwind data owners, and run two full dry-run migrations with automated reconciliation reports before the production load.

Testing combines unit and contract tests in the build pipeline, a 1,200 requests per second load test against production-sized data, an independent penetration test, and two cycles of user acceptance testing supported by scripted scenarios agreed with Northwind process owners. Defects are classified by severity and no severity-1 or severity-2 defect may remain open at go-live.

- Entry criteria: environments ready, test data approved, scripts baselined;
- Exit criteria: all milestone tests passed, defects triaged, sponsor sign-off;

- Evidence: test reports and reconciliation logs retained for the audit file.

8. Delivery Plan, Team and Milestones

Delivery runs for 28 weeks in four phases: discovery and design (weeks 1-5), core build (weeks 6-16), integration and security hardening (weeks 17-24), and pilot and go-live (weeks 25-28). The team of 11 comprises a delivery lead, a solution architect, six engineers, a security engineer and a QA lead.

Table 3 - Milestones

Milestone	Week	Acceptance evidence
M1 Design sign-off	5	Approved architecture and data model
M2 Core workflows demo	12	Working onboarding flow in staging
M3 Integrations complete	22	Passing contract tests for all connectors
M4 Penetration test closed	25	No open critical or high findings
M5 Production go-live	28	Signed go-live checklist

9. Commercial Proposal

Pricing is a fixed fee for a three-year term, itemised below.

Table 4 - Fixed price

Item	Price (USD)
Platform licence (3-year term)	540,000
Implementation services	420,000
Integrations (ERP, TMS, SSO)	210,000
Security hardening and penetration test	90,000
Training	40,000
Contingency reserve	120,000
Total	**1,420,000**

- Prices are in USD, exclusive of taxes, and fixed for the 3-year term.
- Northwind provides timely access to ERP and TMS sandbox environments.
- Up to three change requests per phase are included.

10. Training and Adoption

Training is delivered in three tracks: administrators (two days, hands-on in staging), procurement users (half a day, workflow-based) and vendors (self-service guides and a recorded walkthrough). A train-the-trainer session equips Northwind staff to onboard new colleagues after go-live.

11. Support and Customer Success

Standard support covers business hours with a one-hour response target for severity-1 incidents; 24x7 coverage is available as a priced option. A named technical account manager is assigned

to Northwind.

12. Experience and References

Apex Systems has delivered nine onboarding or supplier-management platforms since 2019. Reference engagements include Harborline Freight (supplier portal, 2023) and Meridian Foods (vendor compliance, 2022). Contact details can be supplied on request.